Machine Learning in Finance Report

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Introduction

Financial markets exhibit complex, non-stationary behavior characterized by alternating periods of bull markets, bear markets, and consolidation phases. Traditional static investment strategies often fail to adapt to these regime shifts, leading to suboptimal risk-adjusted returns. This study proposes a dynamic, regime-aware machine learning framework for equity allocation, addressing the critical need for adaptive strategies in quantitative finance.

The proposed framework consists of three main components. First, we use Gaussian Mixture Models (GMM) to perform unsupervised clustering of 20-day rolling market features, segmenting the time series into three regimes: Steady Bull, Volatile Bear, and Consolidation. Second, we introduce Variational Autoencoders (VAEs) to generate synthetic regime-specific data, which enhances the stability of regime boundaries and reduces noise-induced misclassifications. Third, we apply regime-adaptive factor selection, choosing different combinations of momentum, volatility, and relative strength factors depending on the prevailing market regime. This approach allows the strategy to align with the varying effectiveness of factors across different market conditions.

Our key contributions include the integration of generative AI into regime modeling, which improves classification robustness and reduces false transitions during high-volatility periods, as well as regime-specific factor testing to identify and apply the most effective factors under different market conditions. Empirical backtesting on DJIA constituent stocks from 2020 to 2025 demonstrates the strategy effectiveness, achieving an 18.4% compound annual growth rate and a Sharpe ratio of 0.99. This performance surpasses the benchmark by 6.7% annually and delivers improved downside protection. In terms of risk, the synthetic-enhanced version shows a lower conditional value at risk (CVaR) of -4.8% compared to -5.4% for the baseline model, and it performs more reliably during stress periods, such as the market turbulence surrounding the 2022 interest rate hikes.

The remainder of this report is organized as follows. Section 2 describes the data preparation and regime classification process. Section 3 outlines the construction of predictive factors and their regime-dependent configuration. Section 4 presents backtesting results, followed by a risk analysis in Section 5. Section 6 concludes with a summary of findings and directions for future research.

1 Data Preparation and Market Regime Classification

1.1 Data Preprocessing and Feature Construction

We begin by collecting historical OHLCV data for all Dow Jones Industrial Average (DJIA) constituents, as well as the DJIA index itself, from Yahoo Finance, covering the five-year period preceding the current date. The raw CSV files are parsed and merged by trading date, resulting in a unified panel of time series data containing both individual stocks and the benchmark index.

To ensure data quality, we apply the following missing value handling and data cleaning procedures. Missing price values are forward-filled (using the last valid observation), followed by backward-filling to handle any remaining gaps. If a stock's `Close` column contains only NaN values, it is excluded from further analysis. In addition, any rows where the closing price is missing are dropped, as the closing price is required to compute returns.Daily log returns are then computed using the adjusted closing price.

Stocks with persistent data integrity issues are excluded during this process. We ensure that all resulting dataframes are non-empty and have datetime-indexed rows.

Price series are normalized using min-max scaling to allow for cross-sectional comparison across stocks with different price levels. Outliers are identified using a 60-day rolling Z-score and are adjusted using a median filtering technique.

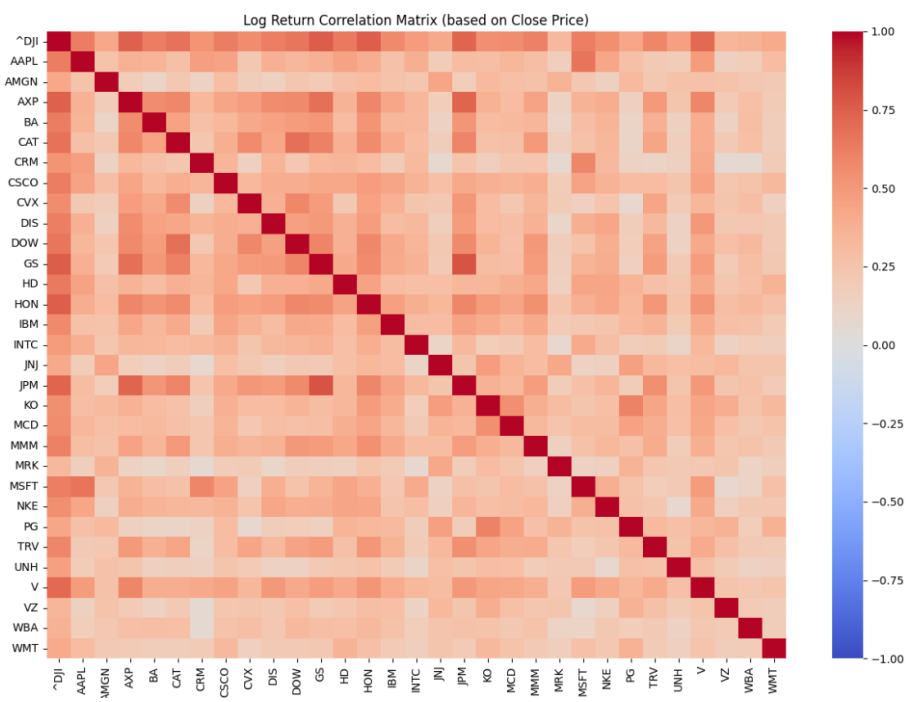

Figure 1: Log return correlation matrix (DJIA)

The correlation matrix heatmap in Figuree 1, constructed using the 30 DJIA constituent stocks and the DJIA index, does not reveal clear sector-level clustering. This suggests that correlations may be influenced by broader market factors rather than strict industry groupings.

1.2 Generalizing Beyond the DJIA

While the main analysis focuses on DJIA constituents, one can use a broader equity dataset, sourced from the TowardAI Whitebox repository for example. After removing stocks with more than 10% NaN values, apply forward and backward filling to handle remaining NaNs. To construct a market index reflective of this wider universe, One can evaluate two stock selection and weighting strategies. The first is a volatility weighted approach, where assets with lower historical volatility received higher weights (inverse vol weighting), promoting stability in the index. The second is based on minimizing average pairwise correlation to ensure that selected stocks contribute less redundant signals. The resulting synthetic index serves as a proxy for market behavior, from which we can derive regime labels using the same unsupervised pipeline as the DJIA based workflow.

Figure 2: Constructed market index using inverse volatility weighting

Figure 3: Constructed market index using minimum correlation stock selection

For regime detection applications, the equal weighted market index (Figure 3) would be more suitable than the volatility weighted one (Figure 2). This is based on several key characteristics that enhance the effectiveness of unsupervised regime identification.

The equal weighted index approach captures clearer demarcations of major market phases: the dramatic bear market decline during the March 2020 crisis, the robust bull market recovery through 2021, and the consolidation patterns observed in 2022. The volatility weighted index, while promoting stability through it inverse volatility weighting, appears to oversmooth these critical regime transitions, potentially obscuring the very market stress periods that are most informative for regime classification.

By avoiding the dampening effects of volatility weighting, the equal weighted index maintains sensitivity to wider market sentiment shifts across the entire stock universe, making it better suited for unsupervised regime detection frameworks.

1.3 Unsupervised Regime Classification via GMM

To capture short-term market behavior, we construct 20-day rolling windows of the following features:

- **avg_mean_log_return_20d**: Average log return over the past 20 days.
- **std_mean_log_return_20d**: Standard deviation of log returns over the past 20 days.
- **avg_volatility_20d**: Average historical volatility over 20 days.
- **positive_return_ratio_20d**: Ratio of days with positive returns in the 20-day window.

These features are standardized and used as inputs for unsupervised clustering. Table 1 2 3 presents the descriptive statistics for each feature under the three GMM-identified regimes.

Table 1: Statistics of GMM Input Features for Regime 0

Feature	Count	Mean	Std	Min	25%	50%	75%	Max	Skew	Kurt
avg_mean_log_return_20d	224	0.001621	0.002606	-0.004859	-0.000102	0.001648	0.003616	0.006743	-0.050	-0.650
std_mean_log_return_20d	224	0.004340	0.000795	0.002636	0.003710	0.004289	0.004966	0.006842	0.393	-0.209
avg_volatility_20d	224	0.021114	0.004932	0.014051	0.017402	0.020629	0.023052	0.035014	0.984	0.806
positive_return_ratio_20d	209	0.542924	0.037967	0.460000	0.511667	0.538333	0.570000	0.640000	0.269	-0.581

Table 2: Statistics of GMM Input Features for Regime 1

Feature	Count	Mean	Std	Min	25%	50%	75%	Max	Skew	Kurt
avg_mean_log_return_20d	408	-0.001562	0.001253	-0.006290	-0.002130	-0.001340	-0.000706	0.000834	-1.192	1.820
std_mean_log_return_20d	408	0.003073	0.000642	0.001601	0.002631	0.002998	0.003464	0.004688	0.411	-0.218
avg_volatility_20d	408	0.015821	0.003137	0.010082	0.013454	0.015190	0.017881	0.025256	0.462	-0.379
positive_return_ratio_20d	408	0.460507	0.031184	0.343333	0.442917	0.465000	0.485000	0.521667	-0.705	0.353

Table 3: Statistics of GMM Input Features for Regime 2

Feature	Count	Mean	Std	Min	25%	50%	75%	Max	Skew	Kurt
avg_mean_log_return_20d	622	0.001274	0.000938	-0.000704	0.000607	0.001146	0.001795	0.004280	0.642	0.084
std_mean_log_return_20d	622	0.002820	0.000571	0.001405	0.002451	0.002839	0.003249	0.004278	-0.120	-0.425
avg_volatility_20d	622	0.014317	0.002062	0.009709	0.012849	0.014038	0.015472	0.021064	0.520	-0.035
positive_return_ratio_20d	622	0.548301	0.032066	0.478333	0.523333	0.545000	0.566250	0.640000	0.606	-0.132

We apply Gaussian Mixture Model (GMM) clustering to the feature windows described above. The model automatically segments the time series into three distinct regimes based on feature similarity, without using any market labels.

To visualize the results, we overlay the assigned regime labels on the DJIA price series. Each segment is color-coded: red for Regime 0, blue for Regime 1, and green for Regime 2, as shown in Figure 4.

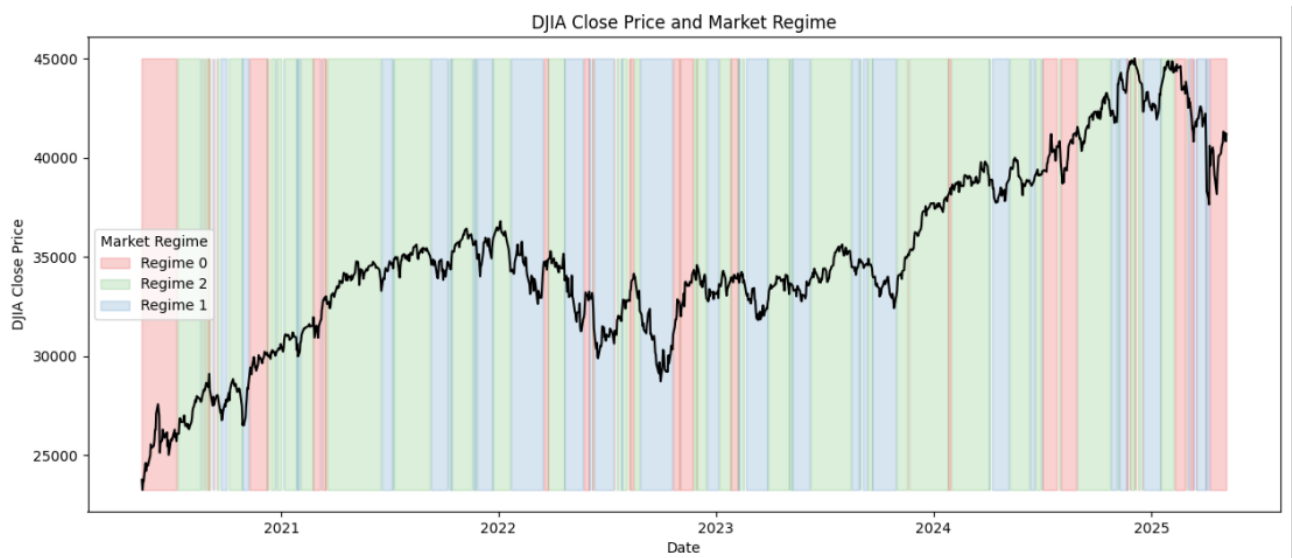

Figure 4: Market regimes over DJIA price

We compare the obtained regime categories with real market data to further refine the definition of market states. This process results in the identification of the following three

market regimes:

- **Regime 0 Steady Bull Market:** Sustained positive returns, low volatility.
- **Regime 1 Volatile Bear Market / Panic:** High volatility, negative returns, rapid drawdowns.
- **Regime 2 Consolidation / Mild Recovery:** Range bound movement with mild upward bias.

Figures 5 67 present the distribution of log returns under each regime. These provide statistical justification for the clustering outcome.

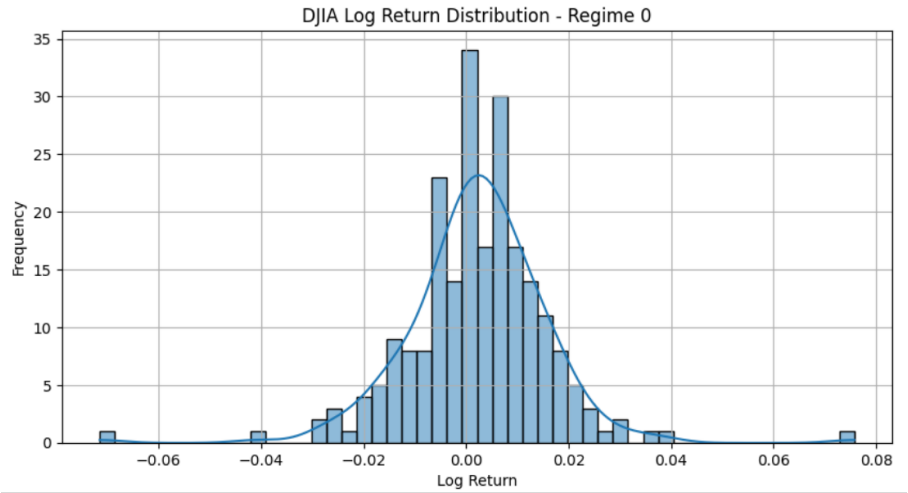

Figure 5: Log returns Regime 0

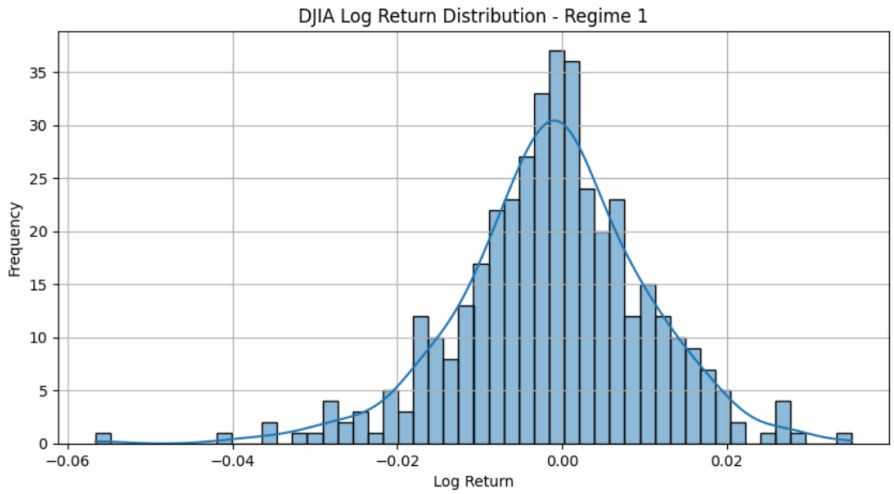

Figure 6: Log returns Regime 1

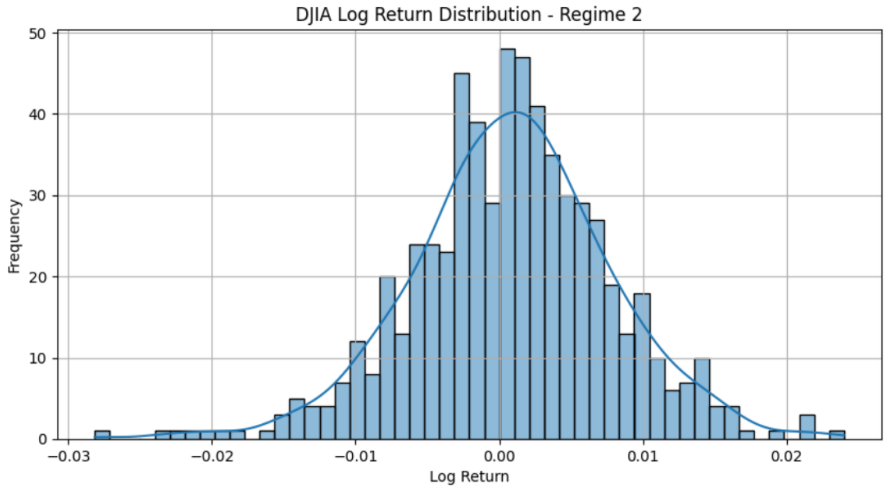

Figure 7: Log returns Regime 2

1.4 Genreated AI and Improved Classification via K-means

For each regime, we train a separate Variational Autoencoder (VAE) to learn and reconstruct the typical feature structure. The encoder compresses inputs into 8-dimensional latent vectors, and the decoder reconstructs the original input. Training minimizes reconstruction error and KL divergence to maintain a standard normal latent distribution. Figures 8-10 show the training history for each regime, illustrating stable convergence.

Figure 8: VAE training Regime 0

Figure 9: VAE training Regime 1

Figure 10: VAE training Regime 2

We then use the decoder to generate synthetic data. Figures 11-13 compare real versus generated feature distributions, demonstrating that the VAE accurately captures regime characteristics.

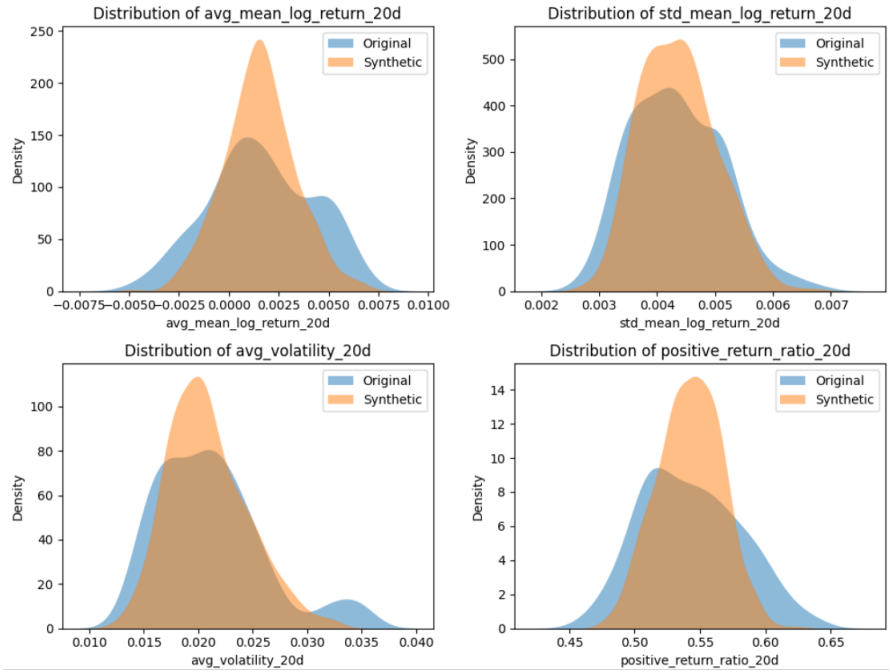

Figure 11: Feature distribution Regime 0

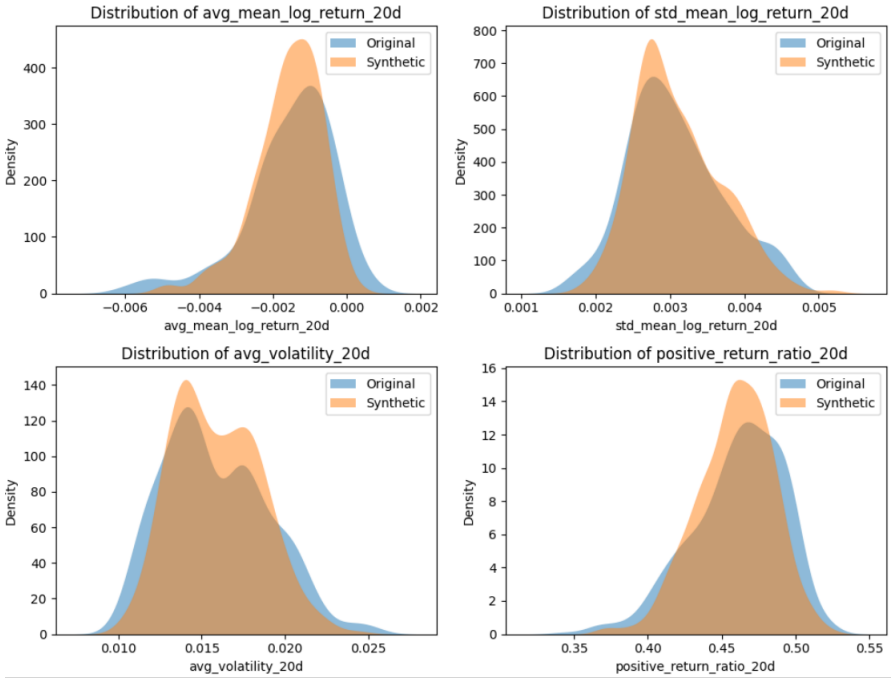

Figure 12: Feature distribution Regime 1

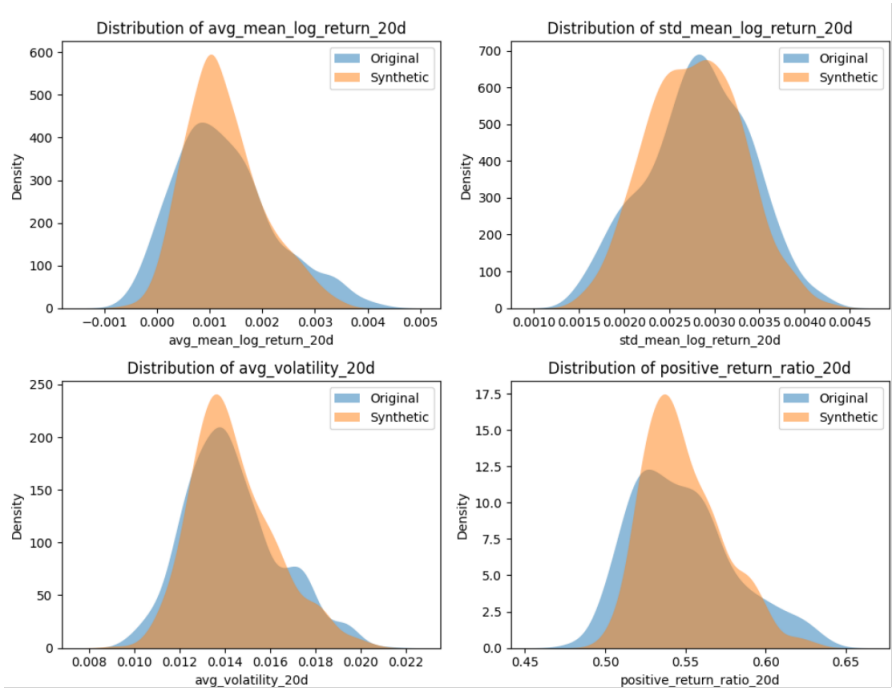

Figure 13: Feature distribution Regime 2

2 Factor Calculation and Testing

2.1 Factor Definitions

This section defines various quantitative factors used in the system. For time-dependent formulas, the variable n represents the lookback window in trading days, with typical values: $n \in \{21\ (1M), 63\ (3M), 126\ (6M), 252\ (12M)\}$.

- **Momentum (MOM_n)**: Price momentum over n trading days:

$$\text{MOM_n}_t = \frac{P_t}{P_{t-n}} - 1 \tag{1}$$

- **Volatility (VOL_n_STD)**: Standard deviation of daily returns over the past n days:

$$\text{VOL_n_STD}_t = \sqrt{\frac{1}{n}\sum_{i=0}^{n-1}(r_{t-i} - \bar{r})^2} \tag{2}$$

  where $\bar{r}$ is the mean return over n days.

- **Relative Strength (RS_n)**: Relative strength of an asset compared to the benchmark index:

$$\text{RS_n}_t = \frac{1 + \text{MOM_n}_t^{\text{asset}}}{1 + \text{MOM_n}_t^{\text{index}}} - 1 \tag{3}$$

- **Sharpe Ratio (Sharpe_n)**: Risk-adjusted return over the past n trading days:

$$\text{Sharpe_n}_t = \frac{\bar{r}_n}{\sigma_n} \tag{4}$$

  where $\bar{r}_n$ is the mean return, and σ_n is the standard deviation over n days.

- **Simple Moving Average (SMA_n)**: Arithmetic mean of prices over n days:

$$\text{SMA_n}_t = \frac{1}{n}\sum_{i=0}^{n-1} P_{t-i} \tag{5}$$

- **Price Relative to SMA (PriceToSMA_n)**: Current price relative to its n-day SMA:

$$\text{PriceToSMA_n}_t = \frac{P_t}{\text{SMA}_n} - 1 \tag{6}$$

2.2 Factor Testing and Regime-Based Configuration

To identify the most effective factors across different market conditions, we conducted a series of empirical tests using quintile-based return analysis. Factors were evaluated based on their ability to differentiate future weekly returns when sorted into quintiles. This testing was carried out within each market regime to reflect regime-specific dynamics.

Based on these results, we configured a regime-aware factor system, where different factors are emphasized depending on the current market regime. Table 4 summarizes the preferred factors and their directional signals for each regime. This configuration provides the foundation for constructing adaptive, regime-sensitive investment strategies.

Table 4: Market Regime Factor Configuration

Regime	**Description**	**Factor**	**Preferred Direction**
0	Steady Bull Market	MOM_12M	Negative
		RS_12M	Negative
		VOL_3M_STD	Positive
		PriceToSMA200	Negative
1	Volatile Bear Market / Panic	MOM_12M	Positive
		RS_12M	Positive
		VOL_3M_STD	Negative
		PriceToSMA50	Positive
2	Consolidation / Mild Recovery	VOL_3M_STD	Negative
		VOL_1M_STD	Negative

3 Trading Strategy and Backtesting

3.1 Framework

We design a dynamic equity allocation strategy based on a regime-switching framework tailored to the Dow Jones Industrial Average (DJIA) index. The core hypothesis is that market conditions evolve through distinct macro regimes, each favoring different factor exposures. By detecting the current regime and dynamically selecting stocks aligned with regime-specific signals, the strategy seeks to outperform a passive DJIA benchmark.

Our approach integrates three data sources:

- **Weekly factor exposures** for DJIA constituents, including momentum, volatility, valuation, and trend metrics.

- **Daily regime classification**, derived from market-wide indicators and labeled into discrete states.
- **Preprocessed daily returns** for all DJIA stocks and the index, used for calculating realized returns and benchmark comparisons.

3.2 Trading Strategy

This strategy is a regime-aware, multi-factor equity selection model targeting the constituents of the Dow Jones Industrial Average (DJIA). The strategy dynamically adjusts its factor exposures based on the prevailing market regime, with the goal of capturing alpha across different macro environments while maintaining robust risk control.

The selection process operates on a weekly basis and involves the following steps:

1. Identify the current market regime from the most recent daily regime data.
2. Extract factor values for all DJIA stocks on the rebalance date.
3. Standardize selected factors using Z-scores, adjusting the sign according to the target direction for the current regime.
4. Combine Z-scores via equal-weighted summation to generate composite scores.
5. Select the top 5 stocks with the highest composite scores.
6. Allocate equal capital to each selected stock.

Table 5 summarizes the key configuration parameters used in the backtesting of the regime-aware factor strategy. These settings reflect practical implementation constraints, such as realistic transaction costs, limited portfolio size, and regular rebalancing intervals. The strategy holds a concentrated portfolio of five DJIA stocks, rebalanced weekly, with regime-specific factor selection and standardized scoring. Transaction costs and turnover are explicitly accounted for, ensuring that reported performance closely reflects real-world trading conditions.

Table 5: Backtesting and Strategy Parameters

Parameter	Value
Initial Capital	$1,000,000
Rebalancing Frequency	Weekly
Number of Stocks Held	5 (equal-weighted)
Transaction Cost	0.10% (10 basis points)
Factor Standardization Method	Z-score
Market Regimes Used	3
Turnover Penalty	Included via transaction cost adjustment

3.3 Backtesting Result with Original Data

We conduct a backtest from 2020 through 2025 using actual historical factor and price data. The overall performance compared to the DJIA index is summarized in Table 6.

Figure 14 15 illustrates the cumulative log return and capital growth of the strategy versus the benchmark.

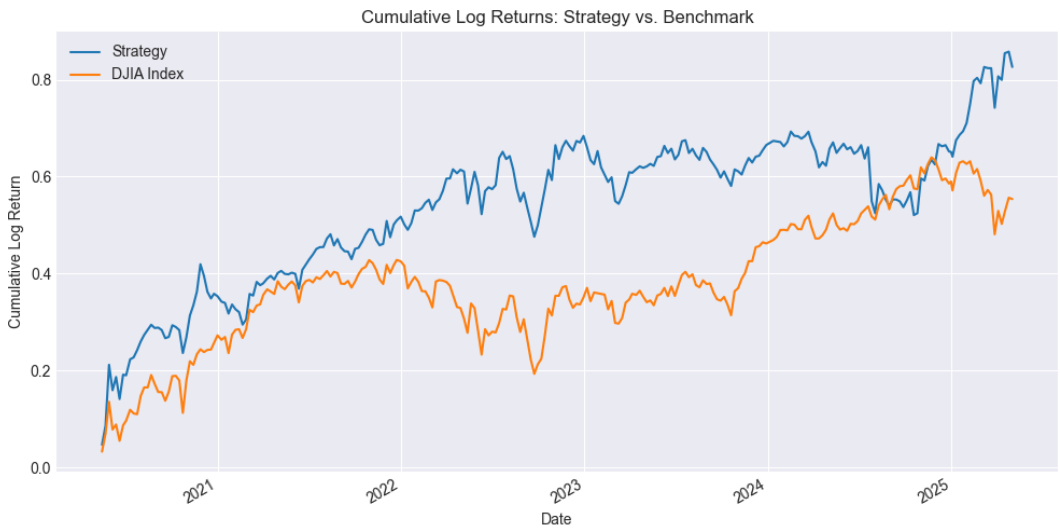

Figure 14: Cumulative Log Returns: Strategy vs. Benchmark

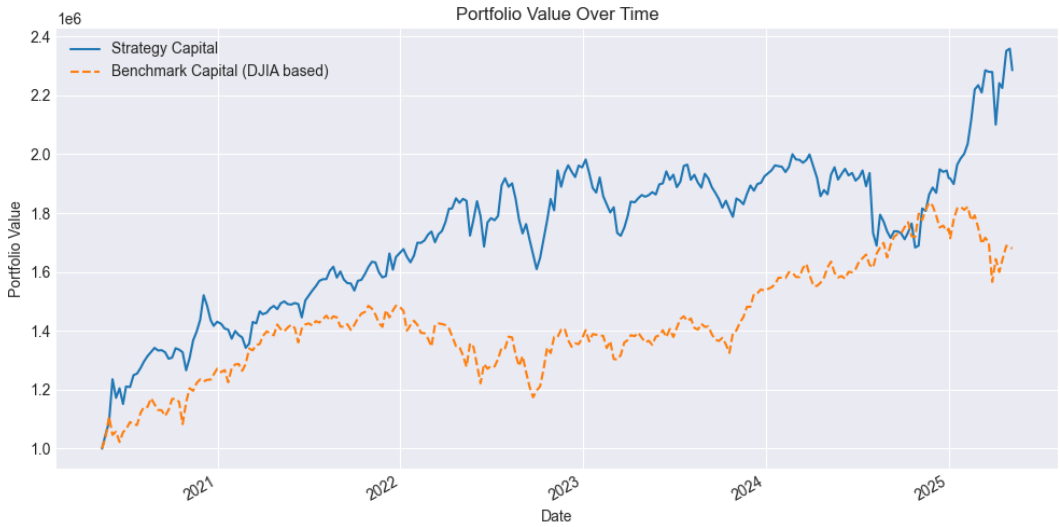

Figure 15: Capital Growth Comparison: Strategy vs. DJIA Benchmark

We further analyze performance under each market regime (see Table 7).

Table 6: Overall Strategy vs. DJIA Benchmark

Metric	Strategy	DJIA Benchmark
Total Return (Simple)	1.2851	0.7396
Annualized Log Return	0.1653	0.1107
Annualized Volatility	0.1840	0.1591
Sharpe Ratio (Log Approx)	0.8981	0.6959
Max Drawdown (Simple)	-0.1609	-0.2095

Table 7: Performance by Market Regime

Regime	Metric	Strategy	Benchmark
Steady Bull (0)	Total Return (Simple)	0.6635	0.3795
	Annualized Log Return	0.5881	0.3718
	Annualized Volatility	0.2971	0.1719
	Sharpe Ratio	1.9791	2.1628
	Max Drawdown	-0.1509	-0.0884
Volatile Bear (1)	Total Return (Simple)	0.3773	0.3722
	Annualized Log Return	0.1958	0.1936
	Annualized Volatility	0.1914	0.1935
	Sharpe Ratio	1.0233	1.0002
	Max Drawdown	-0.1579	-0.1767
Consolidation (2)	Total Return (Simple)	-0.0026	-0.0810
	Annualized Log Return	-0.0010	-0.0338
	Annualized Volatility	0.1109	0.1242
	Sharpe Ratio	-0.0093	-0.2722
	Max Drawdown	-0.1611	-0.2135

The strategy demonstrates superior performance in bear and consolidation phases and matches the benchmark during bull runs, though with a lower Sharpe ratio in bull markets due to higher volatility.

3.4 Backtesting Result with Synthetic Data

To test the robustness of the regime-switching strategy, we introduce a synthetically enhanced regime labeling mechanism. This version incorporates additional simulated data to reinforce regime boundaries, aiming to improve signal stability and portfolio performance.

The backtest using this Synth-Enhanced regime shows that the strategy continues to outperform the DJIA benchmark, with improved Sharpe ratio and reduced drawdown. Table 8 summarizes the overall performance.

Table 8: Overall Performance with Synth-Enhanced Regime

Metric	Strategy	DJIA Benchmark
Total Return (Simple)	1.3081	0.7396
Annualized Log Return	0.1673	0.1107
Annualized Volatility	0.1693	0.1591
Sharpe Ratio (Log Approx)	0.9879	0.6959
Max Drawdown (Simple)	-0.1491	-0.2095

To provide further insight into the return dynamics and capital growth under the Synth-Enhanced regime, we plot the cumulative log returns and the portfolio value over time.

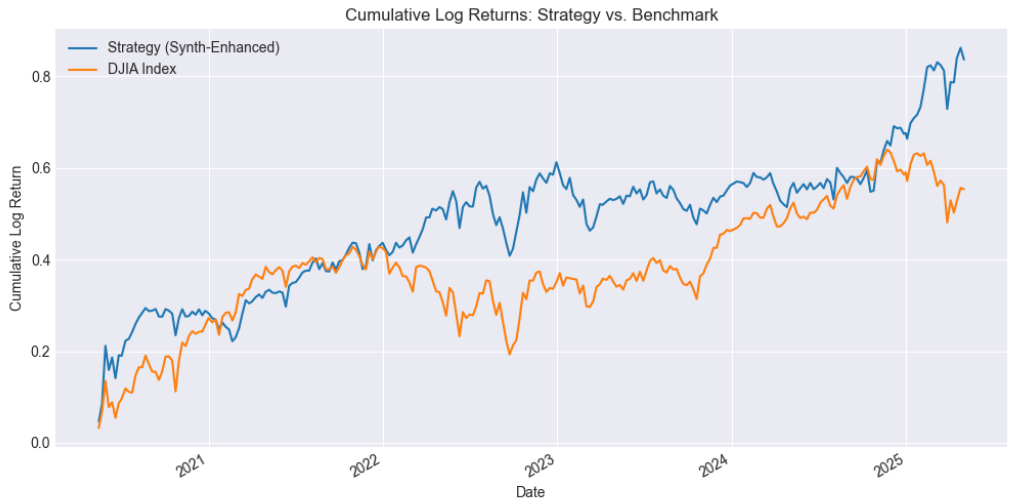

Figure 16: Cumulative Log Returns: Strategy (Synth-Enhanced) vs. DJIA Benchmark

Figure 16 shows that the Synth-Enhanced strategy maintains consistent performance over time and steadily outperforms the benchmark, especially during reversal and low-volatility regimes.

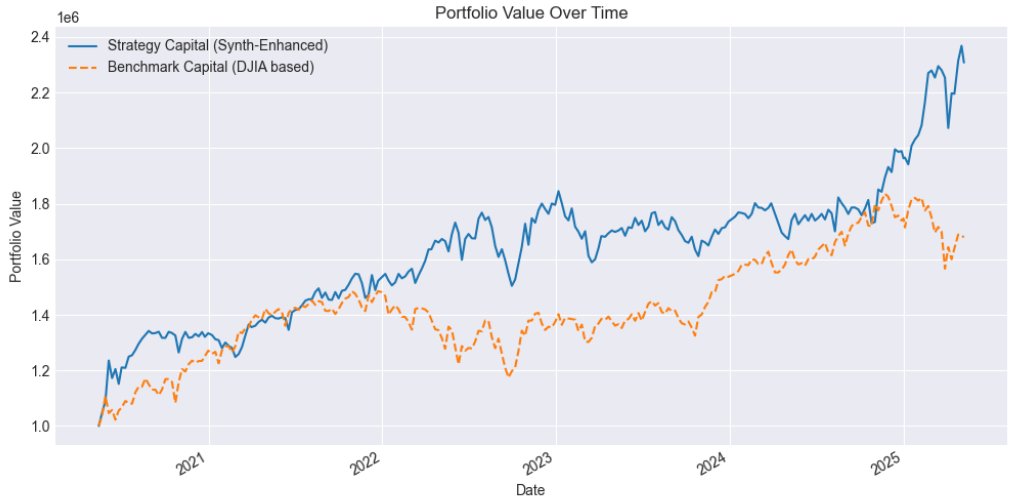

Figure 17: Capital Growth Over Time: Strategy (Synth-Enhanced) vs. Benchmark

Table 9: Performance by Synth-Enhanced Regime

Regime	Metric	Strategy	Benchmark
Regime 0	Total Return (Simple)	0.3021	0.3722
	Annualized Log Return	0.1615	0.1936
	Annualized Volatility	0.1853	0.1935
	Sharpe Ratio	0.8714	1.0002
	Max Drawdown	-0.1183	-0.1767
Regime 1	Total Return (Simple)	0.7736	0.3795
	Annualized Log Return	0.6622	0.3718
	Annualized Volatility	0.2465	0.1719
	Sharpe Ratio	2.6863	2.1628
	Max Drawdown	-0.0682	-0.0884
Regime 2	Total Return (Simple)	-0.0006	-0.0810
	Annualized Log Return	-0.0002	-0.0338
	Annualized Volatility	0.1109	0.1242
	Sharpe Ratio	-0.0021	-0.2722
	Max Drawdown	-0.1591	-0.2135

Figure 17 highlights the effect of compounding returns. The Synth-Enhanced strategy

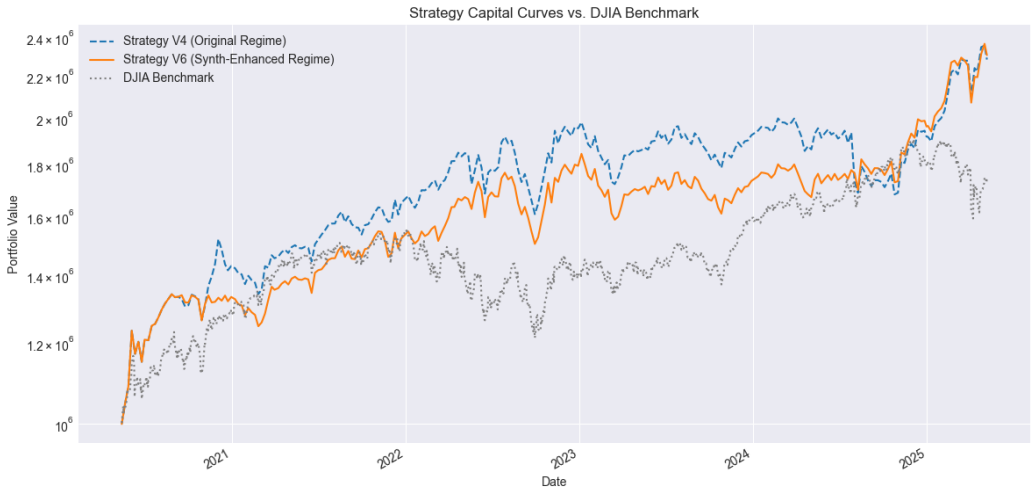

Figure 18: Capital Curves Comparison: Strategy (Original Regime), Strategy (Synth-Enhanced), and DJIA Benchmark

achieves a higher terminal value than the benchmark, while managing volatility more effectively and avoiding severe drawdowns seen in the benchmark during crisis periods.

The performance across the three synthetic regimes is detailed in Table 9. Notably, the strategy exhibits exceptional performance in Regime 1 (reversal phase), with a Sharpe ratio exceeding 2.6.

As shown in Figure 18, both regime-based strategies outperform the DJIA benchmark over the testing period. The original regime strategy exhibits higher peak capital growth but also suffers from larger drawdowns in mid-2022 and late-2023. In contrast, the Synth-Enhanced strategy offers smoother performance and better drawdown control, particularly in volatile and recovery phases. Despite a slightly more conservative growth profile early on, the Regime-Aware dynamic strategy eventually converges with and surpasses the benchmark by a significant margin, supporting its improved robustness under regime-aware design.

4 Risk of Strategy

This section summarizes the key risk metrics used to evaluate the performance of two trading strategies: Original Regime Strategy (Strategy O) and Synth-Enhanced Regime Strategy (Strategy S). The metrics were computed based on weekly capital curves obtained from historical backtesting, as well as market regime classifications.

4.1 Methodology and Risk Metrics

The risk analysis relies on the result files corresponding to Strategy O and Strategy S respectively containing capital curves and regime labels. The following metrics were used to quantify the risk and return profile of each strategy:

- **CAGR (Compound Annual Growth Rate)**: Measures the annualized rate of return assuming compounding. A higher CAGR indicates more effective capital growth over time.
- **Volatility**: Annualized standard deviation of log returns, used to quantify risk. Higher volatility generally implies higher uncertainty.
- **Sharpe Ratio**: The ratio of average return to return volatility, annualized. It reflects the risk-adjusted return; higher values are preferred.
- **Maximum Drawdown**: The largest observed loss from a peak to a trough of a portfolio, before a new peak is attained. It represents downside risk.
- **Skewness and Kurtosis**: Used in tail risk analysis to evaluate asymmetry and tail thickness of return distributions, respectively.
- **VaR (Value-at-Risk)** and **CVaR (Conditional VaR)**: Estimated at a 95% confidence level. VaR gives a loss threshold not exceeded with high probability; CVaR estimates expected losses in the worst-case tail.

4.2 Summary of Results

The computed metrics for each strategy, including per-regime statistics and stress-period analysis, are summarized below:

- **Overall Performance Comparison:**
  - Strategy O - Final Capital: 2,285,061.99, CAGR: 17.97%, Sharpe Ratio: 89.81%
  - Strategy S - Final Capital: 2,322,945.39, CAGR: 18.36%, Sharpe Ratio: 99.54%
- **Regime-wise Risk Metrics:**

  Risk metrics were evaluated separately under each market regime label, capturing the behavior of the strategies in varying conditions (e.g., Neutral/Bullish/Bearish/markets, we define Regime 0,1,2).

  Strategy O (Original Regime):

- Regime 0 (45 weeks): CAGR = 146.27%, Volatility = 29.22%, Sharpe Ratio = 252.58%, Max Drawdown = -13.80%
- Regime 1 (85 weeks): CAGR = 31.99%, Volatility = 18.35%, Sharpe Ratio = -145.18%, Max Drawdown = -14.19%
- Regime 2 (130 weeks): CAGR = 27.84%, Volatility = 11.07%, Sharpe Ratio = 225.23%, Max Drawdown = -14.44%

Strategy S (Synth-Enhanced Regime):

- Regime 0 (84 weeks): CAGR = 31.85%, Volatility = 17.65%, Sharpe Ratio = -181.01%, Max Drawdown = -13.15%
- Regime 1 (45 weeks): CAGR = 151.00%, Volatility = 24.79%, Sharpe Ratio = 280.62%, Max Drawdown = -6.82%
- Regime 2 (131 weeks): CAGR = 28.94%, Volatility = 10.62%, Sharpe Ratio = 282.91%, Max Drawdown = -11.16%

- **Stress Period Performance:** Capital performance was analyzed during specific high-volatility periods:

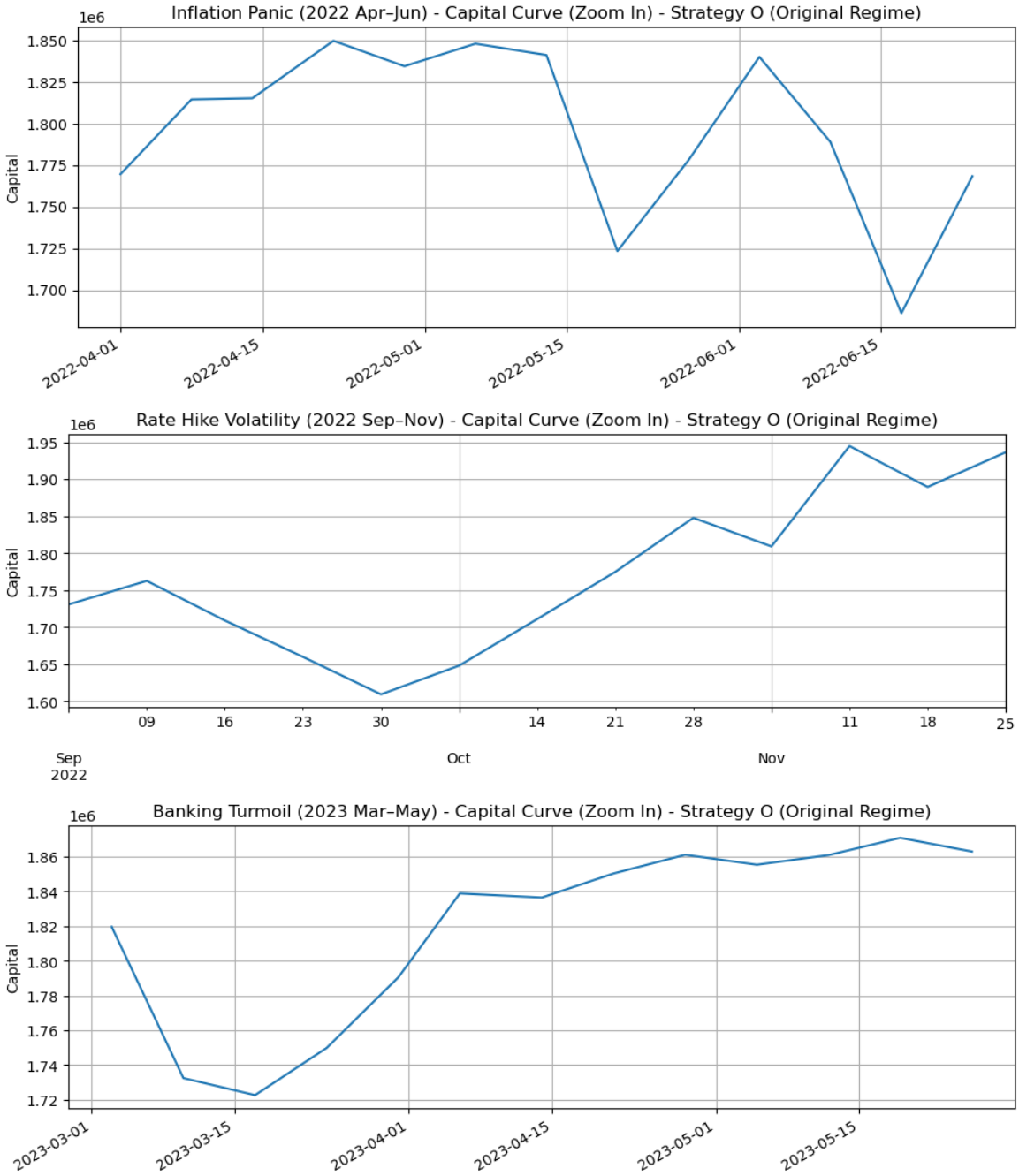

Figure 19: Strategy O - Stress Period Capital Curves

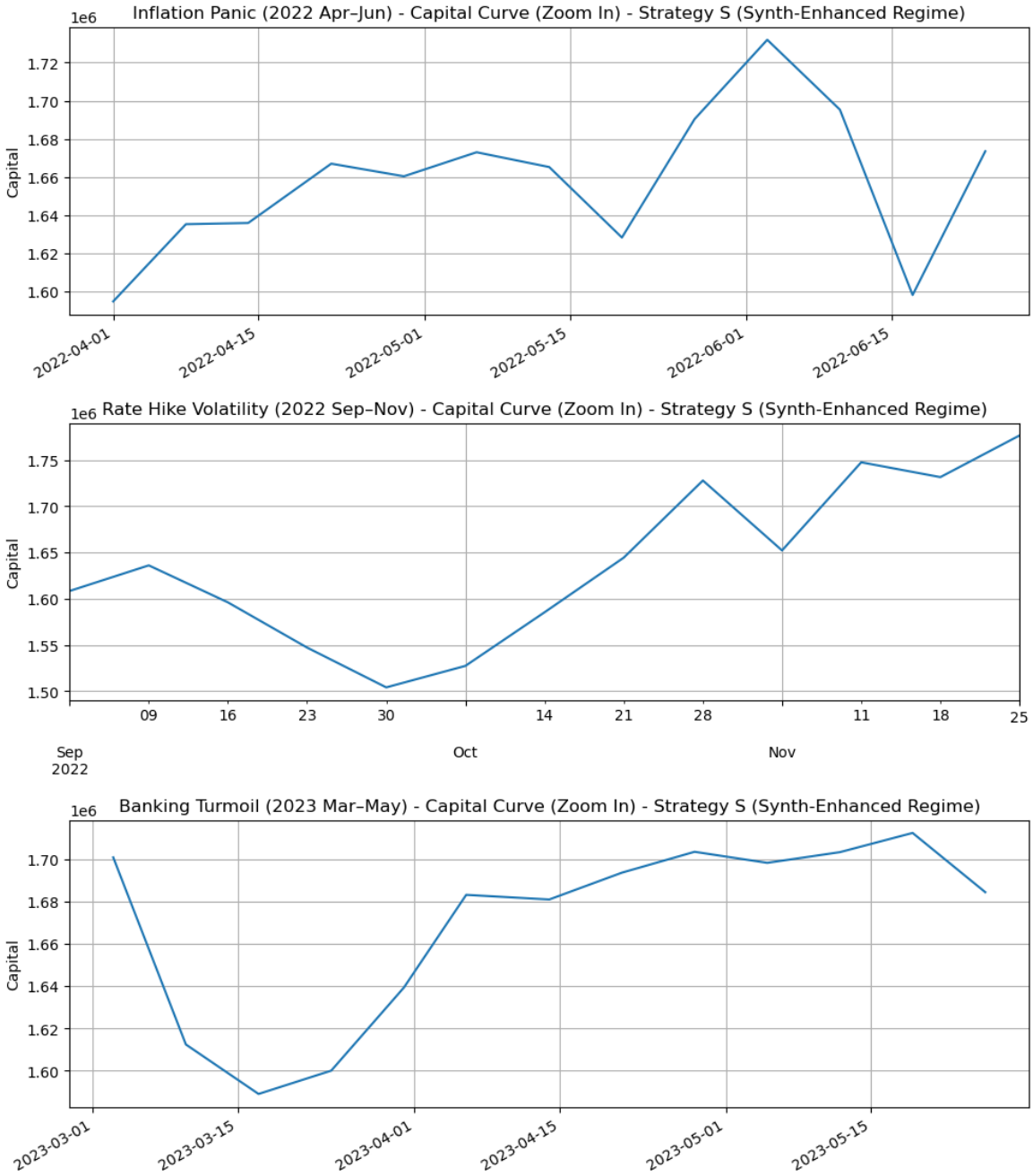

Figure 20: Strategy S - Stress Period Capital Curves

- **Inflation Panic (2022 Apr–Jun)**
  - Strategy O: Return = -0.07%, Max Drawdown = -8.85%
  - Strategy S: Return = 4.94%, Max Drawdown = -7.73%

- **Rate Hike Volatility (2022 Sep–Nov)**
  - Strategy O: Return = 11.86%, Max Drawdown = -8.69%
  - Strategy S: Return = 10.45%, Max Drawdown = -8.06%
- **Banking Turmoil (2023 Mar–May)**
  - Strategy O: Return = 2.38%, Max Drawdown = -5.33%
  - Strategy S: Return = -0.97%, Max Drawdown = -6.58%

- **Tail Risk Analysis:** The weekly return distributions were moderately right-skewed with kurtosis above 3 for both strategies, indicating mild asymmetry and heavier-than-normal tails. This implies potential for significant losses in extreme scenarios.

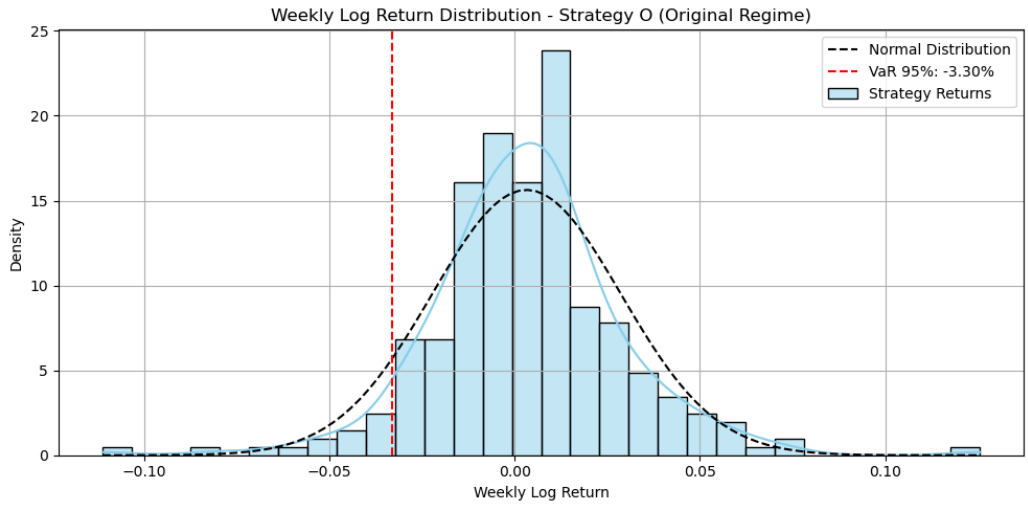

Figure 21: Strategy O - Weekly Log Return Distribution

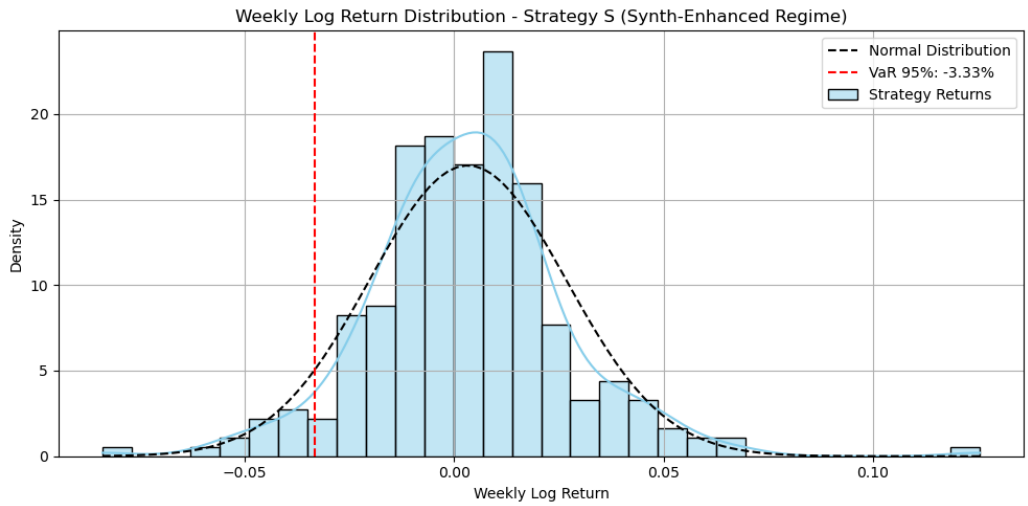

Figure 22: Strategy S - Weekly Log Return Distribution

Value-at-Risk (VaR) and Conditional VaR (CVaR) at 95% level:

- Strategy O: VaR = -3.30%, CVaR = -5.38%
- Strategy S: VaR = -3.33%, CVaR = -4.77%

4.3 Evaluation and Commentary

Overall, both strategies exhibit strong long-term performance with robust growth and moderate risk. Strategy S slightly outperforms Strategy O in terms of final capital and risk-adjusted returns, as evidenced by its higher Sharpe Ratio (99.54%) and slightly lower maximum drawdown (vs). This indicates a more stable return profile under the enhanced regime. While both strategies experience similar drawdown structures, the capital curves suggest that S benefits more during recovery phases.

When analyzing regime-dependent behavior, we observe that both strategies perform exceptionally well in Regime 1 and Regime 2, as reflected by high CAGR and Sharpe Ratios. However, Strategy S demonstrates significantly better stability in adverse regimes: particularly in Regime 1, its drawdown is substantially lower and the Sharpe Ratio more favorable. This suggests that the Synth-Enhanced approach may provide better adaptability in turbulent market conditions.

Stress testing reveals that Strategy S shows more resilience during periods of inflation and interest rate hikes, with lower drawdowns and positive returns. However, during the banking turmoil period, Strategy O maintains a slight positive return while S records a minor loss, indicating possible sensitivity to financial sector-specific shocks.

Tail risk analysis indicates that both strategies are exposed to downside risk, with Strategy O having a marginally higher CVaR. This suggests that extreme negative returns could be slightly more severe for O. Strategy S shows a more symmetric return distribution and a slightly more favorable tail profile.

In conclusion, Strategy S demonstrates stronger overall performance and greater robustness across regimes and extreme market conditions.

5 Conclusion

In conclusion, this study presents a regime-aware quantitative trading framework that combines unsupervised market state detection, generative data augmentation, and adaptive factor selection. By leveraging synthetic data to stabilize regime classification and selecting factors based on their effectiveness in each market condition, the system demonstrates improved performance, robustness, and adaptability. Empirical results confirm that the proposed approach delivers higher risk-adjusted returns and better tail risk control compared to traditional methods. These findings highlight the potential of integrating machine learning and dynamic factor strategies in navigating the complexity of modern financial markets.

References

[1] Sinolink Securities Quantitative Research Group. (2020). Can FinGPT extract trading signals from forum sentiment? â Mining sentiment factors from CSI 300 sectors. *GF Securities Financial Engineering Research Report*.

[2] Sinolink Securities Quantitative Research Group. (2022). ChatGLM-based sentiment stock selection in the medical sector â A practical guide to large language models. *GF Securities Financial Engineering Research Report*.

[3] Sinolink Securities Quantitative Research Group. (2023). Sentiment enhancement strategy for CSI 1000 based on BERT-TextCNN. *GF Securities Financial Engineering Research Report*.

[4] Wang, Z.Y., and Zhang, J. (2023). Generate earthquake catalog using the VAE method. In *Proceedings of the 2nd Asia Conference on Algorithms, Computing and Machine Learning (CACML 2023)*, pp. 283–286.

[5] Rivera, M. (2024). How to train your VAE. In *Proceedings of the 2024 IEEE International Conference on Image Processing (ICIP)*, pp. 3882–3888.